

# Jonathan Lozano-Mayo

*PhD Candidate in Theoretical Physics*

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## RESEARCH INTERESTS

Condensed matter physics · Many-particle physics · Quantum phenomena  
Tensor networks for strongly correlated systems · Non-perturbative phenomena  
Topological and non-topological solitons

## EDUCATION

- 2023–present **Ph.D. in Physics**, *The University of Texas at Austin*, Weinberg Institute for Theoretical Physics  
Advised by Antonios Alvertis.
- 2016–2021 **B.Sc. in Physics**, *National Autonomous University of Mexico (UNAM)*, Faculty of Sciences  
Advised by Manuel Torres Labansat.  
*Thesis*: Spontaneous symmetry breaking and extended field configurations in a scalar theory subject to a potential with two families of vacuum states. **HONORIFIC MENTION**

## PUBLICATIONS

### *Refereed journal articles*

- 2026 **J. Lozano-Mayo**, *Generalized virial identities: radial constraints for solitons, instantons, and bounces*, *J. High Energ. Phys.* **04** (2026) 058 [[arXiv:2512.23548](https://arxiv.org/abs/2512.23548)].
- 2026 **J. Lozano-Mayo** and M. Torres-Labansat, *Multi-kinks and composite oscillons in a commensurable and non-degenerate double sine-Gordon model*, *J. High Energ. Phys.* **09** (2026) 218 [[arXiv:2407.09682](https://arxiv.org/abs/2407.09682)].
- 2021 **J. Lozano-Mayo** and M. Torres-Labansat, *Kink solutions in a generalized scalar  $\phi_G^4$  field model*, *J. Phys. Commun.* **5** (2021) 025004.

### *Under review*

- 2025 J. J. Ziegler, K. Freese, **J. Lozano**, G. Montefalcone, *Explaining the “too massive” high-redshift galaxies in JWST data: numerical study of three effects and a simple relation* [[arXiv:2507.21409](https://arxiv.org/abs/2507.21409)].  
In revision at MNRAS.

### *Collaboration papers*

- 2024–2025 Member of the ALICE Collaboration. Papers carrying my name under ALICE’s collective authorship policy are not listed individually.

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## RESEARCH EXPERIENCE

- 2026–present **Excitons and phonons in strongly correlated materials**, *UT Austin*  
*With Antonios Alvertis.* Computing exciton binding energies in strongly coupled electron-phonon systems with tensor-network methods, in the regime where perturbative treatments of the coupling break down.
- 2025–2026 **Generalized virial identities**, *UT Austin*, **INDEPENDENT PROJECT**  
Derived a one-parameter family of virial identities for  $O(n)$ -symmetric solitons, instantons and bounces that resolves Derrick's global constraint into core and tail contributions. Verified them analytically for the Fubini-Lipatov instanton, the BPS monopole and the BPST instanton, and established their residuals as an accuracy diagnostic for numerical solutions.
- 2024–2025 **Charm production across color environments**, *ALICE Collaboration, UT Austin*  
*With Christina Markert.* Measured  $h$ – $D^0$  azimuthal correlations in ALICE Run-3 Pb–Pb data to probe charm production in different color environments, and contributed performance studies of the upgraded Inner Tracking System (ITS2) as collaboration service work.
- 2024 **High-redshift galaxies and top-heavy initial mass functions**, *UT Austin*  
*With Katherine Freese.* Modeled high-redshift galaxy spectral energy distributions with the population synthesis code PÉGASE to determine where JWST observations remain consistent with  $\Lambda$ CDM.  
**UNDER REVIEW**
- 2020–2022 **Solitons in scalar field theories with non-degenerate vacua**, *UNAM*  
*With Manuel Torres Labansat.* Established the existence of static  $n$ -kink structures in deformed potentials by asymptotic analysis, characterized the inter-kink binding forces numerically, and derived an analytical relation for the multi-kink energy. Renormalized the one-loop quantum mass correction of kinks in a generalized  $\phi^4$  potential. **TWO REFEREED PAPERS**

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## AWARDS & DISTINCTIONS

- 2024 **Biedenharn Endowment for Excellence**, Department of Physics, UT Austin.
- 2022 **Travel Grant**, German Physical Society (DPG) and International Association of Physics Students (IAPS), to compete in the PLANCKS international final at LMU Munich.
- 2021–2022 **Two-time Winner, Mexican Tournament of Physics** (*Torneo Mexicano de Física*), the national theoretical physics competition for bachelor's and master's students that serves as the Mexican PLANCKS preliminary. **1st of 30 teams** in 2022; the 2021 win placed me on the first Mexican team to compete at PLANCKS.
- 2021 **PLANCKS International Final, Porto**: **7th of 50 teams** in Mexico's first appearance at the international theoretical physics competition.
- 2021 **Juan Manuel Lozano Mejía Diploma**, UNAM Institute of Physics: institute-wide distinction for the most outstanding undergraduate thesis research in physics.
- 2021 **PRISMA Cluster of Excellence Fellowship**, Johannes Gutenberg University Mainz.
- 2021 **Honorific Mention**, B.Sc. thesis defense, UNAM: the highest distinction a thesis committee can award.
- 2019 **Research Assistantship**, UNAM Institute of Physics.
- 2017–2018 **Summer research and school scholarships**: Mexican Academy of Sciences Research Summer (2018); UNAM Experimental Physics School (2018); UNAM Physics School (2017).

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## PRESENTATIONS

### *Talks at UT Austin*

Apr 2026 *Exploration of electron-phonon interactions in strongly correlated materials.* Qualifying examination, Department of Physics.

### *Contributed talks*

2025  *$h$ - $D^0$  angular correlations and  $\Delta$ ROF performance studies at ITS2.* ALICE-USA Collaboration Meeting.

2021 *Multi-solitones en teorías escalares de campo con vacíos no-degenerados: el modelo doble de sine-Gordon.* LXVI National Physics Congress, Mexican Physical Society.

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## TEACHING

2023–2026 **Teaching Assistant, *The University of Texas at Austin***

Graduate Quantum Mechanics (2026); undergraduate Waves and Optics (2025), Modern Physics (2024–2025) and Electromagnetism Laboratory, PHY 105N (2023). Led discussion and laboratory sections, gave short pre-lab lectures, graded coursework.

2025 **Lecturer, Quantum Field Theory I, *ICTP Physics Without Frontiers***

Introductory quantum field theory course for the ICTP initiative in Latin America.

2024 **Teaching Assistant, Advanced Particle Physics, *ICTP Physics Without Frontiers***

Graduate course on advanced quantum field theory; led problem-set discussions.

2020–2022 **Teaching Assistant, UNAM, *Faculty of Sciences***

Undergraduate Thermodynamics (2022) and Nuclear and Subnuclear Physics (2020). Ran problem-solving sessions; wrote and graded homework and exams.

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## SERVICE & OUTREACH

2022 Referee, *Journal of Physics G: Nuclear and Particle Physics*.

2017 Volunteer instructor in physics and mathematics, UNAM university preparation course.

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## SKILLS, TRAINING & LANGUAGES

Numerical Tensor networks (DMRG, TDVP), exact diagonalization, finite-difference PDE solvers

Programming Julia, Python, C++, Mathematica, ROOT/O<sup>2</sup> (ALICE), L<sup>A</sup>T<sub>E</sub>X

Schools CMB Data Analysis (UT Austin, 2025); Modeling and Tools for Data Analysis in Science (UANL, 2022); Bad Honnef Physics School on Effective and Lattice Field Theory (DPG, 2021); Mexican Schools on String Theory and Supersymmetry and on Astro-Particle Physics (Guanajuato, 2021)

Languages Spanish (native), English (C<sub>1</sub>), German (A<sub>2</sub>), French (A<sub>2</sub>)